

# HEALTHCARE ANALYTICS

## End-to-End Data Analysis Project Report

Python • MySQL • Power BI

**Prepared by: Ankit Kumar Rai**

Data Analyst | Power BI • SQL • Python

Berlin, Germany

[github.com/ankitrai84](https://github.com/ankitrai84) | [linkedin.com/in/ankitrai19061998](https://linkedin.com/in/ankitrai19061998)

## 1. Executive Summary

---

This project demonstrates a complete, end-to-end data analytics workflow built on a healthcare dataset of 55,500 patient records sourced from Kaggle. The objective was to simulate a real-world analyst pipeline — starting from raw, unclean data and ending with a decision-ready, AI-enhanced Power BI dashboard suitable for executive reporting.

The workflow spans three stages: data cleaning and feature engineering in Python, structured querying and business-question analysis in MySQL, and interactive visualization and storytelling in Power BI. The final deliverable is a 3-page, drill-down enabled dashboard covering patient demographics, clinical outcomes, and financial performance.

|                                 |                                           |                                        |                                    |
|---------------------------------|-------------------------------------------|----------------------------------------|------------------------------------|
| <b>55,500</b><br>Total Patients | <b>\$1.4B+</b><br>Total Revenue (2019-24) | <b>15.5 days</b><br>Avg Length of Stay | <b>33.4%</b><br>Abnormal Test Rate |
|---------------------------------|-------------------------------------------|----------------------------------------|------------------------------------|

## 2. Project Objective

---

The key objectives of this project were to:

- Clean and transform a real-world, messy healthcare dataset into an analysis-ready format
- Engineer new features (Length of Stay, Age Group, Admission Year/Month) to enable deeper analysis
- Use SQL to answer specific business and clinical questions on patient, billing, and insurance data
- Design an interactive, multi-page Power BI dashboard with AI-powered visuals for executive decision-making
- Showcase a complete, portfolio-ready data analytics pipeline from raw data to business insight

### 3. Dataset Overview

Source: Kaggle — Healthcare Dataset | Records: 55,500 | Period: FY 2019 - 2024

The raw dataset contained the following fields:

| Column                             | Description                                                      |
|------------------------------------|------------------------------------------------------------------|
| Name, Age, Gender, Blood Type      | Patient demographic details                                      |
| Medical Condition                  | Diagnosed condition (Cancer, Diabetes, Asthma, etc.)             |
| Date of Admission / Discharge Date | Admission and discharge timestamps                               |
| Doctor, Hospital                   | Treating doctor and hospital name                                |
| Insurance Provider                 | Insurer name (Cigna, Medicare, Aetna, etc.)                      |
| Billing Amount                     | Total billed amount for the visit                                |
| Room Number, Admission Type        | Room assigned and admission category (Elective/Emergency/Urgent) |
| Medication, Test Results           | Prescribed medication and diagnostic outcome                     |

### 4. Methodology

#### 4.1 Data Cleaning & Feature Engineering (Python)

Tools: Pandas, NumPy, Matplotlib, Seaborn (Jupyter Notebook)

- Loaded raw CSV and inspected shape, datatypes, and null values
- Standardized inconsistent text casing in the Name column
- Converted Date of Admission and Discharge Date to proper datetime format
- Identified and removed duplicate records
- Engineered new columns: Length of Stay (days), Age Group (binned), Admission Year, Admission Month
- Produced exploratory visualizations (bar charts, pie charts, heatmaps) to validate data patterns before moving to SQL

### Libraries Importing

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

print("All libraries are Loaded!")
```

All libraries are Loaded!

### Dataset Loading\*

```
[2]: df = pd.read_csv(r"C:\Users\ankit\Downloads\Python Project\Healthcare\healthcare_dataset.csv")

print("Dataset is Loaded")
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")
```

Dataset is Loaded  
Rows: 55500, Columns: 15

### Dataset First View

```
[5]: print(df.isnull().sum())
```

```
Name          0
Age            0
Gender         0
Blood Type     0
Medical Condition  0
Date of Admission  0
Doctor         0
Hospital       0
Insurance Provider  0
Billing Amount  0
Room Number    0
Admission Type  0
Discharge Date  0
Medication     0
Test Results    0
dtype: int64
```

### BASIC STATISTICS

```
[6]: print(df.describe())
```

|       | Age          | Billing Amount | Room Number  |
|-------|--------------|----------------|--------------|
| count | 55500.000000 | 55500.000000   | 55500.000000 |
| mean  | 51.539459    | 25539.316097   | 301.134829   |
| std   | 19.602454    | 14211.454431   | 115.243069   |
| min   | 13.000000    | -2008.492140   | 101.000000   |
| 25%   | 35.000000    | 13241.224655   | 202.000000   |
| 50%   | 52.000000    | 25538.069380   | 302.000000   |
| 75%   | 68.000000    | 37820.508432   | 401.000000   |
| max   | 89.000000    | 52764.276740   | 500.000000   |

```
0]: # NEW COLUMN 2: Age Group -----

df['Age_Group'] = pd.cut(
    df['Age'],

    bins = [0, 18, 35, 50, 65, 100],
    labels = ['0-18', '19-35', '36-50', '51-65', '65+']
)

print("✅ Age Group column created!")
print(df['Age_Group'].value_counts())

✅ Age Group column created!
Age_Group
65+      16096
19-35    13519
51-65    12298
36-50    12167
0-18       886
Name: count, dtype: int64

1]: # NEW COLUMN 3: Year or Month -----

df['Admission_Year'] = df['Date_of_Admission'].dt.year
df['Admission_Month'] = df['Date_of_Admission'].dt.month_name()

print("✅ Year and Month columns created!")
print("Years:", sorted(df['Admission_Year'].unique()))

✅ Year and Month columns created!
Years: [np.int32(2019), np.int32(2020), np.int32(2021), np.int32(2022), np.int32(2023), np.int32(2024)]
```

## 4.2 SQL Analysis (MySQL Workbench)

The cleaned dataset was loaded into MySQL, where structured queries were written to answer specific business questions, including:

- Patients by medical condition
- Average billing amount per medical condition
- Gender-wise patient count and average age
- Insurance provider-wise revenue
- Admission type breakdown
- Age group-wise analysis
- Top 10 doctors by number of patients treated

Example query:

```
SELECT
    `Medical Condition`,
    COUNT(*) AS Total_Patients
FROM healthcare_db.patients
GROUP BY `Medical Condition`
ORDER BY Total_Patients DESC;
```

```
17      -- Query 2 – Average Billing Amount for every condition
18 •    SELECT Medical_Condition,
19         ROUND(AVG(Billing_Amount), 2) AS Avg_Billing,
20         ROUND(MIN(Billing_Amount), 2) AS Min_Billing,
21         ROUND(MAX(Billing_Amount), 2) AS Max_Billing
22     FROM healthcare
23     GROUP BY Medical_Condition
24     ORDER BY Avg_Billing DESC;
25
```

| Result Grid     Filter Rows:   Export:    Wrap Cell Content:  |                   |             |             |             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|-------------|-------------|
|                                                                                                                                                                                                                                                                                                                                                                                                         | Medical_Condition | Avg_Billing | Min_Billing | Max_Billing |
| ►                                                                                                                                                                                                                                                                                                                                                                                                       | Obesity           | 25804.36    | -1310.27    | 52024.73    |
|                                                                                                                                                                                                                                                                                                                                                                                                         | Diabetes          | 25660.48    | -1316.62    | 52211.85    |
|                                                                                                                                                                                                                                                                                                                                                                                                         | Asthma            | 25633.46    | -1520.42    | 52181.84    |
|                                                                                                                                                                                                                                                                                                                                                                                                         | Arthritis         | 25511.78    | -1130       | 52170.04    |
|                                                                                                                                                                                                                                                                                                                                                                                                         | Hypertension      | 25503.06    | -1660.01    | 52764.28    |
|                                                                                                                                                                                                                                                                                                                                                                                                         | Cancer            | 25152.32    | -2008.49    | 52373.03    |

50 -- Query 5 – Admission Type breakdown  
 51  
 52 • SELECT Admission\_Type,  
 53 COUNT(\*) AS Total\_Cases,  
 54 ROUND(AVG(Billing\_Amount), 2) AS Avg\_Billing,  
 55 ROUND(AVG(Length\_of\_Stay), 1) AS Avg\_Stay\_Days  
 56 FROM healthcare  
 57 GROUP BY Admission\_Type  
 58 ORDER BY Total\_Cases DESC;  
 59  
 60

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [fA](#)

|   | Admission_Type | Total_Cases | Avg_Billing | Avg_Stay_Days |
|---|----------------|-------------|-------------|---------------|
| ▶ | Elective       | 18473       | 25612.14    | 15.5          |
|   | Urgent         | 18391       | 25514.53    | 15.4          |
|   | Emergency      | 18102       | 25505.33    | 15.6          |

90 -- Query 8 – Test Results and Billing relation  
 91  
 92 • SELECT Test\_Results,  
 93 COUNT(\*) AS Total\_Patients,  
 94 ROUND(AVG(Billing\_Amount), 2) AS Avg\_Billing,  
 95 ROUND(AVG(Length\_of\_Stay), 1) AS Avg\_Stay\_Days  
 96 FROM healthcare  
 97 GROUP BY Test\_Results  
 98 ORDER BY Avg\_Billing DESC;  
 99  
 100

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [fA](#)

|   | Test_Results | Total_Patients | Avg_Billing | Avg_Stay_Days |
|---|--------------|----------------|-------------|---------------|
| ▶ | Inconclusive | 18198          | 25637.34    | 15.5          |
|   | Abnormal     | 18437          | 25544.04    | 15.5          |
|   | Normal       | 18331          | 25452.22    | 15.5          |

```

176 -- Query 15 - Insurance vs Condition - Most costly combination
177 • SELECT Insurance_Provider, Medical_Condition,
178       COUNT(*) AS Total_Cases,
179       ROUND(AVG(Billing_Amount), 2) AS Avg_Billing,
180       ROUND(SUM(Billing_Amount), 2) AS Total_Billing
181 FROM healthcare
182 GROUP BY Insurance_Provider, Medical_Condition
183 ORDER BY Avg_Billing DESC LIMIT 15;

```

| Insurance_Provider | Medical_Condition | Total_Cases | Avg_Billing | Total_Billing |
|--------------------|-------------------|-------------|-------------|---------------|
| Cigna              | Obesity           | 1847        | 26158.57    | 48314880.24   |
| Blue Cross         | Obesity           | 1872        | 26079.99    | 48821743.56   |
| Aetna              | Hypertension      | 1862        | 25873.86    | 48177132.06   |
| Aetna              | Asthma            | 1723        | 25849.52    | 44538728.67   |
| Medicare           | Hypertension      | 1823        | 25839.63    | 47105654.33   |
| Blue Cross         | Diabetes          | 1846        | 25826.32    | 47675380.49   |
| Medicare           | Obesity           | 1839        | 25817.38    | 47478155.09   |
| Blue Cross         | Arthritis         | 1834        | 25799.66    | 47316578.84   |
| UnitedHealthcare   | Asthma            | 1854        | 25793.31    | 47820788.34   |
| Medicare           | Asthma            | 1812        | 25775.45    | 46705113.26   |
| Medicare           | Diabetes          | 1885        | 25711.12    | 48465464.44   |

Result 11 x

### 4.3 Dashboard Development (Power BI)

MySQL Workbench was connected directly to Power BI Desktop as a live data source. A 3-page interactive dashboard was then built with the following features:

- 3-page layout: Executive Overview, Clinical Analysis, Financial Insights
- Custom page navigation buttons for seamless switching between pages
- Interactive slicers for Gender, Admission Type, and Medical Condition
- Dynamic titles that update automatically based on slicer selection
- Tooltip mini-dashboards showing extra context on hover
- AI Key Influencers visual to identify drivers of abnormal test results
- Decomposition Tree for drill-down revenue analysis by insurer and condition
- Smart Narrative / AI-generated natural language insights
- 15+ custom DAX measures powering KPIs across all three pages
- Drill-down enabled charts on trend and category visuals



## 5. Dashboard Walkthrough

### 5.1 Page 1 — Executive Overview

Provides a high-level summary of patient volume, revenue, average length of stay, and admission trends, along with AI-generated narrative insights.

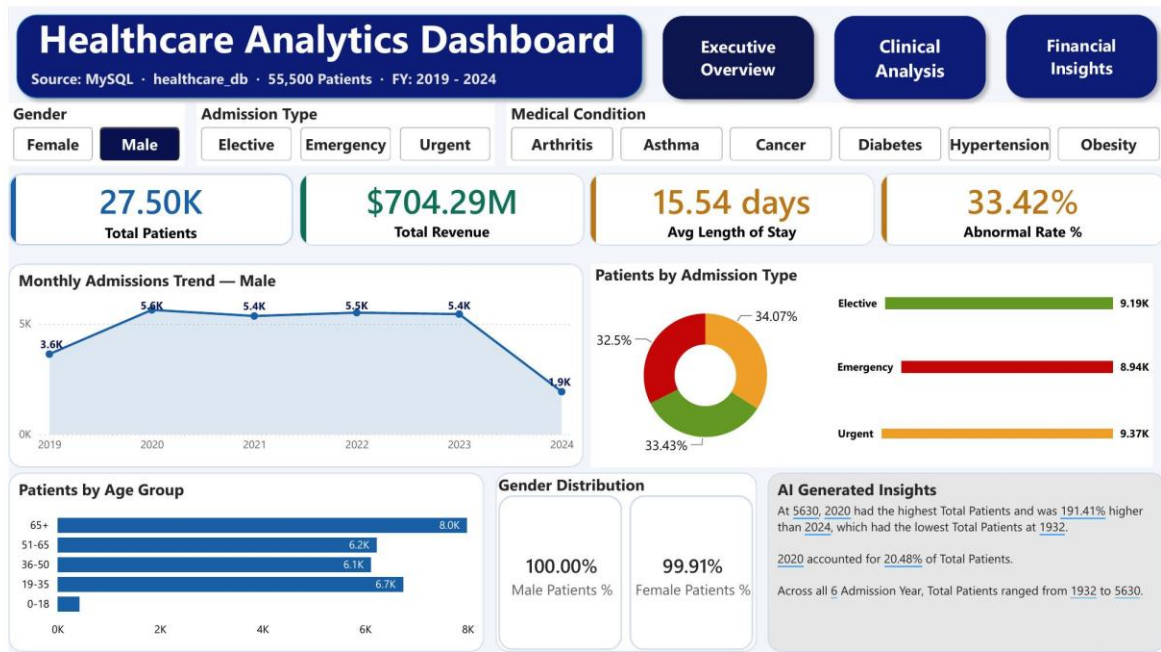


Figure 1: Executive Overview page — KPIs, admission trend, age group and gender distribution

## 5.2 Page 2 — Clinical Analysis

Focuses on medical conditions, test result outcomes, prescribed medications, and blood type distribution, plus an AI Key Influencers visual identifying what drives abnormal test results.

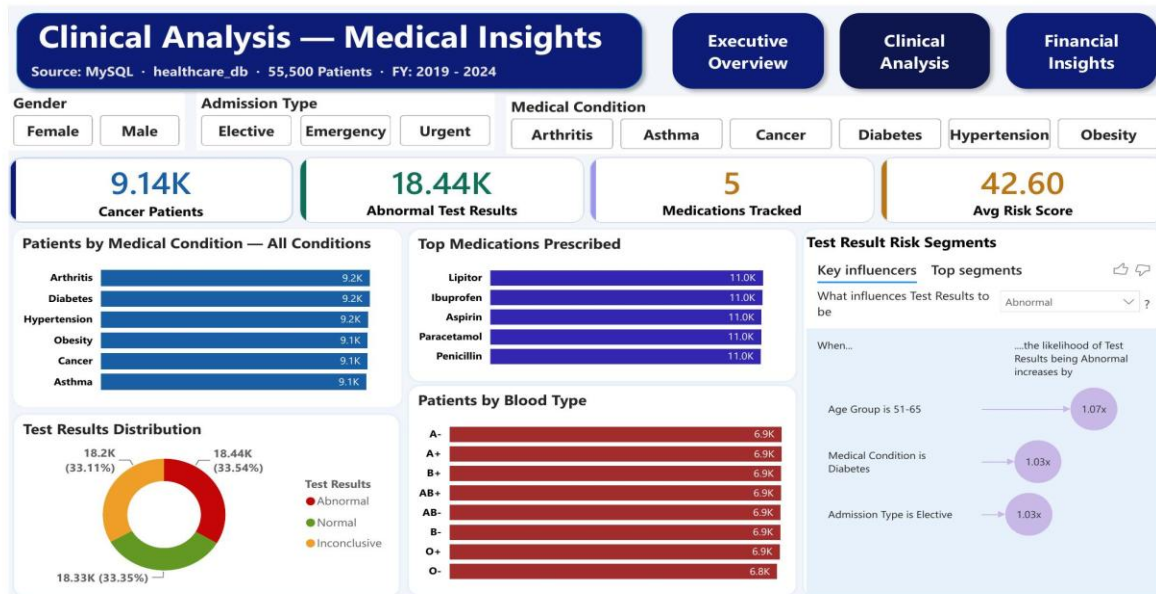


Figure 2: Clinical Analysis page — condition breakdown, test results, key influencers

## 5.3 Page 3 — Financial Insights

Covers revenue performance by insurance provider and medical condition, monthly revenue trends, and a Decomposition Tree for interactive revenue drill-down.

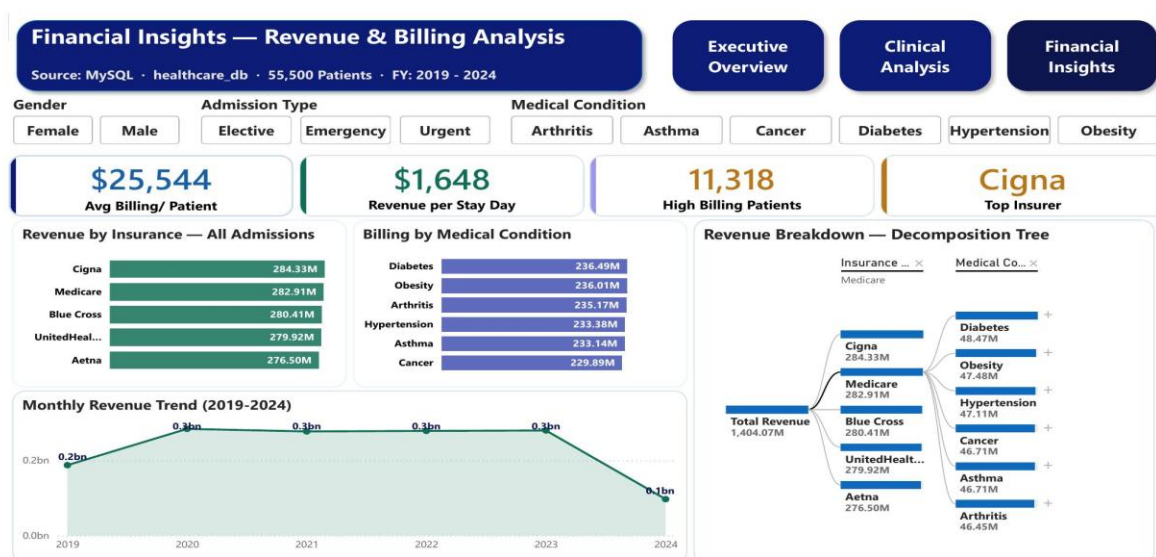


Figure 3: Financial Insights page — revenue by insurer, billing by condition, decomposition tree

## 6. Key Insights

---

- Total patient volume across FY2019-2024 stands at 55,500, generating over \$1.4B in total revenue
- 2020 recorded the highest admissions (5,630), followed by a steady year-on-year decline through 2024
- Patients in the 51-65 age group show a 1.07x higher likelihood of abnormal test results — the strongest single influencer identified by the AI Key Influencers visual
- Cigna is the top-performing insurance provider by total revenue (\$284.33M), narrowly ahead of Medicare and Blue Cross
- Revenue is fairly evenly distributed across medical conditions, with Diabetes generating the highest billing at \$236.49M
- Test results are almost evenly split into three categories — Abnormal (33.5%), Normal (33.4%), and Inconclusive (33.1%) — suggesting no single condition dominates diagnostic risk
- Emergency, Elective, and Urgent admissions are near-evenly distributed (~33% each), indicating balanced hospital resource demand across admission types

## 7. Tech Stack & Project Links

---

| Layer               | Tools Used                                      |
|---------------------|-------------------------------------------------|
| Data Source         | Kaggle (Healthcare Dataset — 55,500 records)    |
| Data Cleaning       | Python — Pandas, NumPy                          |
| Visualization (EDA) | Matplotlib, Seaborn                             |
| Database            | MySQL Workbench                                 |
| BI & Dashboarding   | Power BI Desktop (Power Query, DAX, AI Visuals) |

**Repository:** [github.com/ankitrai84/Healthcare-Analytics-Dashboard](https://github.com/ankitrai84/Healthcare-Analytics-Dashboard)

**LinkedIn:** [linkedin.com/in/ankitrai19061998](https://www.linkedin.com/in/ankitrai19061998)

## 8. Conclusion

---

This project reflects a complete, self-driven data analytics workflow — from raw data ingestion to executive-ready business intelligence. It demonstrates practical proficiency across the modern analytics stack: Python for data preparation, SQL for structured business analysis, and Power BI for advanced, AI-assisted visualization and storytelling.

The techniques applied here — feature engineering, relational querying, DAX measures, drill-down interactivity, and AI-powered insight generation — mirror the responsibilities of a real-world Data Analyst / BI Developer role, making this project a strong, hands-on demonstration of end-to-end analytical capability.